

HUSSAIN AHMAD

PhD Applicant — Intelligent URLLC Wireless Communication

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GitHub



PROFILE

Research-oriented engineer with an MSc from the University of Southern Denmark and experience in wireless communication systems, AI-enabled networking, and resilient system design. Strong background in Python- and MATLAB-based simulation, network modeling, and performance evaluation for low-latency and ultra-reliable communication scenarios, including 5G/6G research and reinforcement learning for communication optimization. Combines knowledge of multi-hop networking, interference, and reliability with hands-on embedded systems and validation experience, and is strongly motivated by resilient communication for drone swarms and defence-relevant systems. **Publications list available on page 2**

EXPERIENCE

Embedded Hardware & Software Developer

Bitzer Electronics

Feb 2022 – Feb 2026

Sønderborg, Denmark

- Built a high-throughput sensor validation system combining hardware, STM32 firmware (C), and Python-based diagnostic scripts for automated data collection and analysis.
- Designed PCBs and managed end-to-end integration (schematics, layout, BOM, production handoff), ensuring reproducible hardware configurations and clear technical documentation.
- Implemented data logging, debugging, and structured test workflows, directly transferable to communication testbeds, structured system validation, and reliable data collection.

Research Assistant

Sejong University

Aug 2019 – Sep 2021

Seoul, South Korea

- Designed and simulated distributed network algorithms in Python, focusing on real-time optimization of resource allocation and latency in edge computing frameworks that are core challenges in 6G and next generation networks.
- Applied Reinforcement Learning to optimize dynamic systems, achieving 33% performance gains—a skill directly applicable to optimizing AI-driven instrumentation workflows.
- Developed a comprehensive end-to-end communication system simulation in MATLAB, designing modulators, channel models, and receivers for a 500 Mbps optical link.
- Conducted detailed numerical simulations of signal propagation and impairment analysis, building a foundational skill set directly applicable to **custom channel modeling** for near-field and sparse array scenarios.

PUBLICATIONS

- Ahmad, H., Aslam, S., Naseem, A. (2026). **Enhancing URLLC in 6G Edge Networks via ICN and Near-Zero Handover**. *IEEE Journal of Internet of Things* (Submitted).
- Ahmad, H., Aslam, S., Naseem, A. (2026). **Semantic-Aware Resource Orchestration for URLLC in 6G mmWave Edge Computing Networks**. *IEEE Open Journal of the Communication Society* (Submitted).
- Ahmad, H., Islam, M. Z., Ali, R., Haider, A., & Kim, H. (2021). AI-enabled Stretch Optimization in Information-Centric Networking for Tactile Internet. *MDPI, Applied Sciences*, 11(16), 7351. [DOI](#)
- Ahmad, H., Islam, M. Z., Ali, R., Haider, A., & Kim, H. (2021). Q learning based Routing in Information-Centric Networking for edge computing 5G framework. *2021 ICGHIT*
- Ahmad, H., Gillani, S. M. T., Omer, T., et al. (2020). Futuristic Short Range Optical Communication: A Survey. *2020 ICISCT*. [DOI](#)

TECHNICAL SKILLS

Programming & Data

Python, MATLAB/Simulink, C/C++, Verilog, Git, Linux, data analysis, simulation workflows, NS3, OMNET++.

Methods

Machine learning, reinforcement learning, quantitative modelling, experimental design.

Engineering

Embedded systems, sensor data pipelines, diagnostics, debugging, PCB design, hardware–software co-design.

Wireless Communication

3GPP, 5G, 6G, mmWave, line-of-sight communications, mobile networks, edge computing, resource allocation, optimization theory, URLLC

EDUCATION

MSc in Electronics Engineering

University of Southern Denmark

2021 – 2023

Sønderborg, Denmark

BSc in Electrical Engineering

Bahria University

2015 – 2019

Islamabad, Pakistan

REFERENCES

Dr. Saleem Aslam

Professor, Bahria University, Pakistan
saleem.buic@bahria.edu.pk

Dr. Rashid Ali

Senior Lecturer, Department of Engineering Science, University West, Sweden
rashid.ali@hv.se

LANGUAGES

- English (professional proficiency)

CORE STRENGTHS

- **Wireless & URLLC/6G:** Experience with low-latency and ultra-reliable communication concepts from research on URLLC in 5G/6G edge networks, including interference, reliability, and tight delay constraints
 - **Simulation & Network Modeling:** Strong skills in Python and MATLAB for link-level and network-level simulations, building custom testbeds, and modeling multi-hop communication behavior under varying channel conditions.
 - **Mesh & Multi-hop Networking:** Background in information-centric networking, routing, and resource orchestration that transfers directly to designing and analyzing mesh and swarm communication schemes in challenging environments.
 - **Resilient & Dependable Systems:** Hands-on experience with validation plans, risk assessment, structured testing, and traceable documentation from quality-focused and regulated settings, aligning with dependable wireless communication for defense and critical applications
 - **Embedded & Testbed Implementation:** Practical expertise in embedded systems, sensor data pipelines, and hardware–software co-design, enabling implementation and evaluation of communication ideas on real platforms, not only in simulations.
 - **Research & Communication:** Track record of peer-reviewed publications, international collaboration, and clear technical writing and presentations, supporting high-quality PhD research and interaction with academic, defense, and industrial partners.
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NOTABLE PROJECTS

- **URLLC-oriented 6G Edge Networking Simulator (Python/MATLAB):** Developed a Python and MATLAB simulation framework for ultra-reliable low-latency communications in 6G edge networks, integrating realistic channel models, traffic patterns, and queueing delays. Implemented and evaluated reinforcement-learning based resource allocation and routing schemes to meet strict latency and reliability targets under interference and traffic fluctuations.
- **Link-level PHY Simulator with Adaptive Modulation and Coding (MATLAB/C++):** Built a link-level simulator for multi-antenna wireless systems implementing modulation, channel coding, and basic MIMO configurations in MATLAB, with selected performance-critical kernels ported to C++ for speed. Investigated BER/BLER performance under AWGN and fading channels, and tested adaptive modulation and coding based on instantaneous channel estimates.
- **FPGA-based Baseband Prototype for Short-Range Communication (VHDL/C):** Designed and implemented a baseband processing chain on an FPGA platform (e.g. Xilinx Spartan series), including modulation/demodulation, filtering and synchronization blocks in VHDL. Integrated the FPGA with an embedded processor in C for control, parameter configuration and data logging, enabling real-time experiments and validation of algorithmic assumptions against hardware constraints.
- **Embedded C/C++ Firmware for Real-Time Sensor Communication (C/C++/Embedded Linux):** Developed C/C++ firmware for a real-time sensor acquisition system with kHz-level sampling, implementing communication protocols, buffering, and low-latency data transfer on microcontroller and embedded Linux platforms. Optimised memory and timing to guarantee deterministic behaviour, and built Python tools for monitoring, analysis, and automated test campaigns.

INTERESTS

- Fitness & Health
- eGaming